SAT Math Mastery · Extra Practice

Thin Skills: Statistics, Probability, Functions and Similarity

Time about 2 hours: Part A in class, Part B independently.

Topics evaluating claims · margin of error · probability · two-variable data · composition and transformations of functions · sums and products of roots · similar triangles and area ratios

Name / Date _____ _____

PART A · IN CLASS — 25 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos (or a calculator), one expression per line; tap a grey dot to read an intersection or a vertex.

1. HARD · NO CALCULATOR NEEDED

A researcher recruited 120 volunteers who responded to an online advertisement. She randomly assigned 60 of them to study with method X and 60 to study with method Y for a month. The method X group had a higher mean score on a final test, and the difference was larger than would be expected from random assignment alone. Which conclusion is best supported by the study?

(A) The study provides no evidence about either method.

(B) Method X caused higher scores for volunteers like those in the study. ✓

(C) Method X is associated with higher scores, but no causal conclusion is possible.

(D) Method X causes higher scores for all students.

1. The 120 volunteers were randomly assigned, 60 to each method. Random assignment is what allows a cause-and-effect conclusion.
2. The difference was larger than random assignment alone would produce, so the difference is real: method X caused the higher scores.
3. But the people were volunteers who answered an advertisement, not a random sample of all students. The conclusion applies only to volunteers like them, so "all students" is too broad, and "no causal conclusion" ignores the random assignment.

Answer: (B)

2. Hard · No calculator needed

A random sample of 500 adults was selected from a large city, and each adult reported how many hours per week they exercise and how many hours they sleep each night. The adults who exercise more tend to sleep more. Which conclusion is best supported by this study?

(A) Exercising more causes adults in the city to sleep more.

(B) Sleeping more causes adults in the city to exercise more.

(C) Among the 500 adults only, exercise and sleep are associated; nothing can be said about the city.

(D) Among adults in the city, exercise and sleep are positively associated, but exercise has not been shown to cause more sleep. ✓

1. The 500 adults were chosen at random from the city, so the result can be generalized to adults in the city.
2. Nobody was assigned to exercise more or less: people just reported what they do. That makes it an observational study, which can show association only, never cause.
3. So the best supported conclusion is a positive association among adults in the city, without a claim that exercise causes more sleep. Choices A and B claim causation; C refuses to generalize even though the sample was random.

Answer: (D)

3. Hard · No calculator needed

In a poll of a random sample of voters, 51% said they support a proposed tax. The margin of error was 2.5% at the 95% confidence level. Based on the poll, which statement is best supported?

(A) A majority of all voters support the tax.

(B) It is plausible that a majority support the tax, but less than half supporting it is also plausible. ✓

(C) Fewer than half of all voters support the tax.

(D) Exactly 51% of all voters support the tax.

1. Plausible values for the population percent run from $51 - 2.5 = 48.5$ to $51 + 2.5 = 53.5$ percent.
2. The interval contains 50%, so both a majority (above 50%) and less than half (below 50%) are plausible.
3. That matches the statement saying a majority is plausible but less than half is also plausible. Statements A and C claim more than the poll shows, and D claims the exact value.

Answer: (B)

4. HARD · NO CALCULATOR NEEDED

A random sample of 500 residents of Town P gave an estimate of 30% for the percent who favor a new park, with a margin of error of 4%. A random sample of 500 residents of Town Q gave an estimate of 38%, also with a margin of error of 4%. Which statement is best supported?

(A) It is plausible that the percent in favor is the same in both towns. ✓

(B) The percent in favor in Town P is exactly 4 points less than that in Town Q.

(C) Town Q definitely has a greater percent in favor than Town P.

(D) It is not possible for the percents to be equal.

1. Town P: 30 ± 4 gives 26 to 34 percent. Town Q: 38 ± 4 gives 34 to 42 percent.
2. The intervals overlap: both contain 34 percent, so the same value for both towns is plausible.
3. So it is plausible that the percent in favor is the same in both towns. The estimates differ, but the margins of error mean that difference is not established.

Answer: (A)

5. HARD · NO CALCULATOR NEEDED

A school has 1,200 seniors and 2,800 students in other grades. A random sample of 300 seniors was surveyed, and an estimated 62% plan to attend college, with a margin of error of 5%. Which statement is best supported by the survey?

(A) Exactly 62% of the seniors at the school plan to attend college.

(B) Between 57% and 67% of the 300 surveyed seniors plan to attend college.

(C) Between 57% and 67% of all students at the school plan to attend college.

(D) It is plausible that between 57% and 67% of the seniors at the school plan to attend college. ✓

1. The interval is 62 ± 5, which gives 57% to 67%.
2. But the sample of 300 was drawn only from the 1,200 seniors, so the interval describes seniors at the school, not all 4,000 students.
3. A margin of error gives plausible values for the population, not an exact value and not the percent of the sample, so only the statement about seniors fits.

Answer: (D)

6. HARD · NO CALCULATOR NEEDED

Events A and B are independent. The probability of A is 0.4, and the probability that at least one of A and B occurs is 0.7. What is the probability of B?

1. For independent events $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$. Let $P(B) = p$.
2. $0.4 + p - 0.4p = 0.7$, so $0.6p = 0.3$.
3. $p = \dfrac{0.3}{0.6} = 0.5$.

Answer: 0.5 **(or** $\dfrac{1}{2}$**)**

7. HARD · NO CALCULATOR NEEDED

Two fair six-sided dice are rolled. Given that at least one of the dice shows an even number, what is the probability that the sum of the two dice is 8?

1. Count all outcomes with at least one even die: $36 - 9 = 27$ (the 9 all-odd outcomes are removed from 36).

2. Sums of 8: $(2,6), (6,2), (3,5), (5,3), (4,4)$. The pairs $(3,5)$ and $(5,3)$ have two odd dice, so they are not in the condition.

3. Three outcomes remain, so the probability is $\frac{3}{27} = \frac{1}{9} \approx 0.1111$.

Answer: $\frac{1}{9}$ **(or** 0.1111**)**

8. HARD · NO CALCULATOR NEEDED

In a town, 30% of the residents are over 60 years old. Of the residents over 60, 80% vote in elections; of the residents who are not over 60, 50% vote. A voter is chosen at random from all residents who vote. What is the probability that the voter is over 60?

1. Voters over 60: $0.3 \times 0.8 = 0.24$ of all residents. Voters not over 60: $0.7 \times 0.5 = 0.35$.

2. All voters: $0.24 + 0.35 = 0.59$ of all residents.

3. Probability: $\frac{0.24}{0.59} = \frac{24}{59} \approx 0.4068$.

Answer: $\frac{24}{59}$ **(or** 0.4067 **or** 0.4068**)**

9. HARD · NO CALCULATOR NEEDED

The table shows values of x and y for a data set.

x	0	1	2	3	4
y	8	12	18	27	40.5

Which equation fits all five pairs?

(A) $y = 8(1.5)^x$ ✓ (B) $y = 4x + 8$

(C) $y = 2x^2 + 2x + 8$ (D) $y = 8(1.2)^x$

1. Look at x up by 1: y goes $8, 12, 18, 27, 40.5$. The differences $(4, 6, 9, 13.5)$ are not constant, so it is not linear and (B) fails.

2. Ratios of consecutive y-values: $\frac{12}{8} = 1.5$, $\frac{18}{12} = 1.5$, $\frac{27}{18} = 1.5$, $\frac{40.5}{27} = 1.5$. The ratio is constant, so the growth is exponential with $b = 1.5$ and $a = 8$.

3. Check (C) at $x = 2$: $2(4) + 2(2) + 8 = 20 \neq 18$, and (D) at $x = 1$: $8(1.2) = 9.6 \neq 12$. Only $y = 8(1.5)^x$ fits all five.

Answer: (A)

10. HARD · DESMOS

Ten data points lie exactly on the line $y = 2x + 1$, at $x = 1, 2, \ldots, 10$. A new data point $(11, 40)$, far above that line, is added. How does the least-squares line for all eleven points compare with the line $y = 2x + 1$?

(A) Smaller slope and greater y-intercept

(B) The same slope and the same y-intercept

(C) Greater slope and smaller y-intercept ✓

(D) Greater slope and greater y-intercept

1. The line $y = 2x + 1$ predicts $2(11) + 1 = 23$ at $x = 11$. The new point is 40, which is 17 above the line, at the far right end.
2. A high point at the right end pulls the right side of the line up. Predict that the slope gets greater, and since the line has to tilt up around the middle of the data, the intercept gets smaller.
3. Check in Desmos: make a table with columns x_1 and y_1 holding the eleven points, then type `y_1~mx_1+b`.
4. It gives $m \approx 2.773$ and $b \approx -2.091$: the slope is greater than 2 and the intercept is smaller than 1.

Answer: (C)

11. HARD · NO CALCULATOR NEEDED

The residuals for the first four of five data points about a least-squares line are -2, 3, 1 and -4. What is the residual for the fifth point?

1. For a least-squares line the residuals add to 0.
2. The first four add to $-2 + 3 + 1 - 4 = -2$.
3. So the fifth residual is $0 - (-2) = 2$.

Answer: 2

12. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 5$. The function g is defined by $g(x) = f(2x)$. What are the x-intercepts of the graph of $y = g(x)$?

(A) $x = -2$ and $x = \dfrac{5}{2}$

(B) $x = -4$ and $x = 10$

(C) $x = -1$ and $x = \dfrac{5}{2}$ ✓

(D) $x = 0$ and $x = 3$

1. $g(x) = f(2x)$ is zero when the input $2x$ is a zero of f, so $2x = -2$ or $2x = 5$.
2. Divide by 2: $x = \dfrac{-2}{2} = -1$ and $x = \dfrac{5}{2} = 2.5$.
3. Replacing x by $2x$ squeezes the graph horizontally by a factor of 2, so the intercepts move toward the y-axis (not out to -4 and 10).

Answer: (C)

13. HARD · NO CALCULATOR NEEDED

The functions f and g are defined for the values in the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	4	1	2	3

What is the value of $g\left(f^{-1}(g(3))\right)$?

1. First $g(3)$: the table gives 2.
2. Next $f^{-1}(2)$: find the x with $f(x) = 2$. The table gives $f(4) = 2$, so $f^{-1}(2) = 4$.
3. Finally $g(4) = 3$.
Answer: 3

14. HARD · NO CALCULATOR NEEDED

The function f is odd, which means $f(-x) = -f(x)$ for every x. Which of the following functions must be even, meaning $h(-x) = h(x)$ for every x?

(A) $[f(x)]^2$ ✓

(B) $f(f(x))$

(C) $f(x) + 1$

(D) $f(x) + x$

1. Test each option by replacing x with $-x$ and using $f(-x) = -f(x)$.
2. (A) $[f(-x)]^2 = [-f(x)]^2 = [f(x)]^2$: unchanged, so it is even. (B) $f(f(-x)) = f(-f(x)) = -f(f(x))$: odd.
3. (C) $f(-x) + 1 = -f(x) + 1$: neither. (D) $f(-x) + (-x) = -(f(x) + x)$: odd. Only (A) must be even.
Answer: (A)

15. HARD · NO CALCULATOR NEEDED

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 - 5$. Which describes all real values of x for which $f(g(x))$ is defined?

(A) $x \geq 1$

(B) $-\sqrt{6} \leq x \leq \sqrt{6}$

(C) all real numbers

(D) $x \leq -\sqrt{6}$ **or** $x \geq \sqrt{6}$ ✓

1. $f(g(x)) = \sqrt{g(x) - 1} = \sqrt{x^2 - 5 - 1} = \sqrt{x^2 - 6}$.
2. It is defined when $x^2 - 6 \geq 0$, so $x^2 \geq 6$.
3. That means $|x| \geq \sqrt{6} \approx 2.449$: $x \leq -\sqrt{6}$ or $x \geq \sqrt{6}$. (Not $-\sqrt{6} \leq x \leq \sqrt{6}$: that is where it is undefined.)
Answer: (D)

16. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ passes through the point $(4, -2)$. Which point must lie on the graph of $y = 3f(x-1) + 5$?

(A) $(5, 11)$ **(B) $(5, -1)$ ✓**

(C) $(3, -1)$ (D) $(3, 11)$

1. $f(x-1)$ shifts the graph right 1, so the point $(4, -2)$ becomes $(5, -2)$.
2. Multiplying by 3 and adding 5 changes the y-value: $3(-2) + 5 = -1$.
3. So the point is $(5, -1)$. Check: $3f(5-1) + 5 = 3f(4) + 5 = 3(-2) + 5 = -1$.

Answer: (B)

17. HARD · NO CALCULATOR NEEDED

The equation $2x^2 - 7x + k = 0$, where k is a constant, has two real solutions r and s with $r^2 + s^2 = 9$. What is the value of k?

1. By Vieta, $r + s = \dfrac{7}{2} = 3.5$ and $rs = \dfrac{k}{2}$.
2. Use $r^2 + s^2 = (r+s)^2 - 2rs$: $9 = 12.25 - k$, because $(3.5)^2 = 12.25$ and $2rs = k$.
3. So $k = 12.25 - 9 = 3.25$. (The discriminant $49 - 8k = 23 > 0$, so the roots are real.)

Answer: $\dfrac{13}{4}$ **(or** 3.25**)**

18. HARD · NO CALCULATOR NEEDED

What is the sum of all real solutions of $2^{2x} - 6 \cdot 2^x + 8 = 0$?

1. Let $u = 2^x$. Then $2^{2x} = u^2$ and the equation becomes $u^2 - 6u + 8 = 0$.
2. Factor: $(u-2)(u-4) = 0$, so $u = 2$ or $u = 4$.
3. Then $2^x = 2$ gives $x = 1$ and $2^x = 4$ gives $x = 2$.
4. The sum of the solutions is $1 + 2 = 3$.

Answer: 3

19. HARD · NO CALCULATOR NEEDED

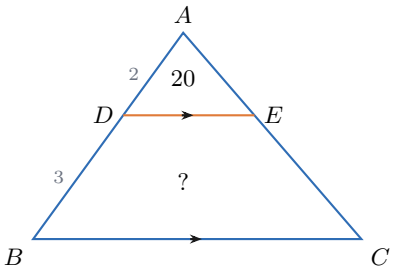
Triangle ABC is similar to triangle DEF. The area of triangle ABC is 36 and the area of triangle DEF is 100. The perimeter of triangle ABC is 24. What is the perimeter of triangle DEF?

1. The ratio of the areas is $\dfrac{100}{36} = \dfrac{25}{9}$, so the scale factor (the square root) is $\dfrac{5}{3}$.
2. Perimeters scale by the scale factor itself, not its square.
3. Perimeter of DEF: $24 \times \dfrac{5}{3} = 40$.

Answer: 40

20. HARD · NO CALCULATOR NEEDED

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD : DB = 2 : 3$ and the area of triangle ADE is 20, what is the area of quadrilateral $DBCE$?



1. $AD : DB = 2 : 3$ means $AD : AB = 2 : 5$, so the small triangle ADE is similar to ABC with scale factor $\frac{2}{5}$.

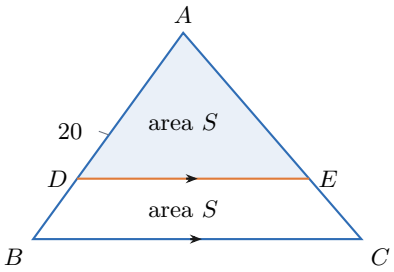
2. Areas scale by the square: $\frac{4}{25}$. So area $ABC = 20 \times \frac{25}{4} = 125$.

3. Quadrilateral $DBCE$ is the rest: $125 - 20 = 105$.

Answer: 105

21. HARD · CALCULATOR

In triangle ABC, point D is on $\overline{AB}$ and point E is on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. The segment $\overline{DE}$ divides the triangle into two regions of equal area. If $AB = 20$, what is the length of $\overline{AD}$, to the nearest tenth?



1. $\triangle ADE \sim \triangle ABC$, and $\triangle ADE$ has half the area of $\triangle ABC$ (the two regions are equal).

2. Area ratio $\frac{1}{2}$, so the length ratio is $\frac{AD}{AB} = \frac{1}{\sqrt{2}}$.

3. Multiply by $AB = 20$: $AD = \frac{20}{\sqrt{2}}$. Type `20/sqrt(2)` : it returns 14.1421, which is 14.1 to the nearest tenth.

Answer: 14.1

22. HARD · NO CALCULATOR NEEDED

A square pyramid has a base with area 144. A plane parallel to the base cuts the pyramid at a point one-third of the way from the apex to the base, forming a smaller square cross-section. What is the area of the cross-section?

1. The cross-section is similar to the base, with the scale factor equal to the ratio of the distances from the apex.

2. The plane is one-third of the way down, so the scale factor is $\frac{1}{3}$ and the area factor is $\frac{1}{9}$.

3. Area: $144 \times \frac{1}{9} = 16$.

Answer: 16

23. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ has x-intercepts at $x = 1$ and $x = 6$. The function g is defined by $g(x) = f(3 - x)$. What is the greatest x-intercept of the graph of $y = g(x)$?

1. $g(x) = f(3 - x)$ equals zero when $3 - x$ is a zero of f: $3 - x = 1$ or $3 - x = 6$.

2. So $x = 3 - 1 = 2$ or $x = 3 - 6 = -3$.

3. The greatest x-intercept is 2.

Answer: 2

24. HARD · NO CALCULATOR NEEDED

The solutions of $x^2 - 5x + 2 = 0$ are r and s. What is the value of $\frac{r}{s} + \frac{s}{r}$?

1. By Vieta, $r + s = 5$ and $rs = 2$.

2. Combine the fractions: $\frac{r}{s} + \frac{s}{r} = \frac{r^2 + s^2}{rs}$, and $r^2 + s^2 = (r + s)^2 - 2rs = 25 - 4 = 21$.

3. So the value is $\frac{21}{2} = 10.5$.

Answer: $\frac{21}{2}$ **(or** 10.5**)**

25. HARD · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 - 7x^2 + 14x - 8$ has three real zeros. What is the sum of the squares of the zeros?

1. By Vieta for the cubic, the zeros have sum 7 and the sum of the products of pairs is 14.

2. Use $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc)$.

3. $49 - 2(14) = 49 - 28 = 21$. (The zeros are $1, 2, 4$, and $1 + 4 + 16 = 21$.)

Answer: 21

PART B · SELF-ASSESSMENT — 20 PROBLEMS

26. HARD · NO CALCULATOR NEEDED

A random sample of students at a university reported a mean of 3.2 hours of study per day, with a margin of error of 0.4 hours. The university has 18,000 students. Based on the sample, what is the greatest total number of study hours per day that is plausible for the whole student body?

1. Plausible values for the mean run from $3.2 - 0.4 = 2.8$ to $3.2 + 0.4 = 3.6$ hours per day.

2. The greatest total comes from the greatest plausible mean.

3. Multiply by the number of students: $3.6 \times 18{,}000 = 64{,}800$ hours per day.

Answer: 64,800

27. HARD · NO CALCULATOR NEEDED

To estimate the number of fish in a lake, a biologist caught and tagged 150 fish and released them. A week later she caught a random sample of 60 fish from the lake, and 9 of them were tagged. Assuming the sample is representative and the tagged fish mixed evenly, what is the best estimate of the number of fish in the lake?

1. The tagged fraction of the sample should match the tagged fraction of the lake: $\dfrac{9}{60} = \dfrac{150}{N}$, where N is the number of fish.

2. The sample fraction is $\dfrac{9}{60} = 0.15$.

3. Solve $0.15\,N = 150$: $N = \dfrac{150}{0.15} = 1{,}000$ fish.

Answer: 1000

28. HARD · NO CALCULATOR NEEDED

A company's customers were asked whether they renewed their contract. The results are shown in the table.

	Renewed	Did not renew
Plan A	84	36
Plan B	45	35

If one of the customers who did not renew is chosen at random, what is the probability that the customer was on Plan B?

1. Only the customers who did not renew count: $36 + 35 = 71$ customers.

2. Of these, 35 were on Plan B, so the probability is $\dfrac{35}{71}$.

3. As a decimal, $\dfrac{35}{71} \approx 0.493$. (Dividing by all 200 customers is the trap.)

Answer: $\dfrac{35}{71}$ **(or** 0.4929 **or** 0.4930**)**

29. HARD · NO CALCULATOR NEEDED

A bag holds 5 red marbles and 3 blue marbles. Two marbles are drawn at random, one after the other, without replacement. What is the probability that the two marbles are the same color?

(A) $\frac{15}{28}$

(B) $\frac{13}{28}$ ✓

(C) $\frac{5}{14}$

(D) $\frac{17}{32}$

1. Both red: $\frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56}$.

2. Both blue: $\frac{3}{8} \cdot \frac{2}{7} = \frac{6}{56}$.

3. Add the two ways: $\frac{26}{56} = \frac{13}{28}$.

Answer: (B)

30. HARD · NO CALCULATOR NEEDED

A line of best fit for a data set is $\hat{y} = 3.2x + 14$. For the data point with $x = 10$, the actual value is $y = 51$. What is the residual for this point (actual value minus predicted value)?

1. Predicted value at $x = 10$: $3.2(10) + 14 = 46$.

2. Residual = actual − predicted $= 51 - 46$.

3. That equals 5 (positive, so the point is above the line).

Answer: 5

31. HARD · NO CALCULATOR NEEDED

A least-squares line for a data set is $\hat{y} = 1.5x + 4$. The mean of the x-values is 10. What is the mean of the y-values?

1. A least-squares line always passes through the point $(\bar{x}, \bar{y})$, the means.

2. Put $\bar{x} = 10$ into the line: $\bar{y} = 1.5(10) + 4$.

3. $\bar{y} = 15 + 4 = 19$.

Answer: 19

32. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = \dfrac{2x+1}{x-3}$ for $x \neq 3$. What is the value of $f^{-1}(5)$?

1. $f^{-1}(5)$ is the input x for which $f(x) = 5$, so there is no need to find the whole inverse function.

2. Solve $\dfrac{2x+1}{x-3} = 5$: $2x + 1 = 5(x-3) = 5x - 15$.

3. Then $16 = 3x$, so $x = \dfrac{16}{3} \approx 5.333$, and $x \neq 3$ is satisfied.

Answer: $\dfrac{16}{3}$ **(or** 5.333**)**

33. HARD · NO CALCULATOR NEEDED

What is the product of all real solutions of $x^4 - 5x^2 + 4 = 0$?

1. Treat it as a quadratic in x^2: $(x^2 - 1)(x^2 - 4) = 0$.

2. So $x^2 = 1$ or $x^2 = 4$, giving $x = \pm 1$ and $x = \pm 2$.

3. The product is $(1)(-1)(2)(-2) = 4$. (Vieta for the whole quartic gives the same: the constant term is 4.)

Answer: 4

34. HARD · NO CALCULATOR NEEDED

A right triangle has legs of length 6 and 8. A larger, similar right triangle has area 150. What is the length of the hypotenuse of the larger triangle?

1. The small triangle is a 6-8-10 right triangle with area $\dfrac{1}{2}(6)(8) = 24$.

2. Area ratio $\dfrac{150}{24} = \dfrac{25}{4}$, so the scale factor is $\dfrac{5}{2} = 2.5$.

3. The hypotenuse 10 scales to $10 \times 2.5 = 25$.

Answer: 25

35. HARD · NO CALCULATOR NEEDED

The graph of the quadratic function f has vertex $(4, -8)$ and passes through the point $(0, 8)$. The function g is defined by $g(x) = f(2x)$. What is the value of $g(1)$?

1. Write the vertex form $f(x) = a(x-4)^2 - 8$.

2. Use $(0, 8)$: $8 = a(0-4)^2 - 8$, so $16a = 16$ and $a = 1$.

3. $g(1) = f(2) = (2-4)^2 - 8 = 4 - 8 = -4$.

Answer: -4

36. HARD · NO CALCULATOR NEEDED

The function f is even, so $f(-x) = f(x)$ for every x. The function g is defined by $g(x) = f(x) + f(-x)$. If $f(3) = 7$ and $f(-5) = 2$, what is the value of $g(3) + g(5)$?

1. Because f is even, $f(-x) = f(x)$, so $g(x) = f(x) + f(-x) = 2f(x)$.
2. $g(3) = 2f(3) = 14$.
3. $f(5) = f(-5) = 2$, so $g(5) = 2(2) = 4$. Total $14 + 4 = 18$.

Answer: 18

37. MEDIUM · NO CALCULATOR NEEDED

A website for a cycling club posted a poll asking, "Should the city spend more on bike lanes?" Of the 1,850 people who chose to respond, 91% voted yes. Which is the main reason the poll result should not be used to estimate the opinion of all city residents?

(A) Respondents selected themselves and are likely to hold similar views. ✓

(B) The sample size is too large.

(C) The question has too few possible answers.

(D) The poll was conducted online, and no online poll can be accurate.

1. The poll was posted on a cycling club's website and people chose whether to answer. That is a voluntary-response sample.
2. People who choose to answer a bike-lane poll on a cycling club's site mostly like bike lanes, which is why 91% of 1,850 responses said yes.
3. A large number of responses does not fix bias from self-selection, so the main problem is that respondents selected themselves.

Answer: (A)

38. MEDIUM · NO CALCULATOR NEEDED

A random sample of 500 adults reported their coffee consumption. The adults who drink coffee daily had a lower rate of a certain heart condition than those who do not. Which change to the study would make it possible to conclude that drinking coffee reduces the rate of the condition?

(A) Increase the sample to 5,000 adults.

(B) Include only adults older than 40.

(C) Randomly assign the participants to drink coffee daily or not, then compare the rates. ✓

(D) Ask each adult about more lifestyle habits.

1. In the study the adults chose for themselves whether to drink coffee, so other differences between the groups (age, lifestyle) could explain the lower rate.
2. A cause-and-effect conclusion needs an experiment: randomly assigning who drinks coffee balances those other factors across the groups.
3. A bigger sample (5,000), restricting by age, or collecting more habits does not remove the problem. Only random assignment does.

Answer: (C)

39. Medium · No calculator needed

Two independent random samples were taken from the same population to estimate a proportion at the same confidence level. The first sample had 400 members and a margin of error of 6%. The second sample had 1,600 members. Which is most likely the margin of error for the second sample?

(A) 1.5% (B) 6%

(C) 3% ✓ (D) 12%

1. The margin of error is proportional to $\frac{1}{\sqrt{n}}$, so only the change in n matters.

2. The sample grows from 400 to 1,600, which is 4 times as large, and $\sqrt{4} = 2$.

3. Divide the margin of error by 2: $\frac{6}{2} = 3$ percent. (Dividing by 4 is the common trap.)

Answer: (C)

40. Medium · No calculator needed

Of 500 residents surveyed, 240 own a car, 180 own a bicycle, and 70 own both. If one of the surveyed residents is chosen at random, what is the probability that the resident owns neither a car nor a bicycle?

1. Own a car or a bicycle: $240 + 180 - 70 = 350$ (subtract the 70 counted twice).

2. Own neither: $500 - 350 = 150$ residents.

3. Probability: $\frac{150}{500} = 0.3$.

Answer: 0.3

41. Medium · No calculator needed

In a trial, 90 of 120 patients given Drug X improved, and 60 of 100 patients given Drug Y improved. How many times as likely is a patient given Drug X to improve as a patient given Drug Y?

1. Drug X: $\frac{90}{120} = 0.75$ improved.

2. Drug Y: $\frac{60}{100} = 0.6$ improved.

3. Compare: $\frac{0.75}{0.6} = 1.25$ times as likely.

Answer: 1.25

42. Medium · No calculator needed

A teacher modeled the final score y of students in a course with the line of best fit $\hat{y} = 82 - 1.6x$, where x is the number of days the student was absent. Which statement is the best interpretation of the number 82 in this model?

(A) A student absent for 82 days is predicted to score 0.

(B) The predicted score decreases by 82 for each day absent.

(C) The mean score of all students in the class is 82.

(D) The predicted score of a student with no absences is 82. ✓

1. In $\hat{y} = 82 - 1.6x$ the number 82 is the constant term, the y-intercept.
2. It is the value of $\hat{y}$ when $x = 0$: $82 - 1.6(0) = 82$.
3. Here x is the number of days absent, so $x = 0$ means no absences: the predicted score of a student with no absences is 82. The slope -1.6 is the change per day, and 82 is not a class mean.

Answer: (D)

43. Medium · No calculator needed

A model $y = 0.85x + 12$ was fitted to data in which x ranged from 20 to 80. A student uses the model to predict y when $x = 150$. Which statement about this prediction is best?

(A) It is reliable, because the slope is positive.

(B) It is less reliable, because 150 is outside the range of the data. ✓

(C) It is impossible, because y cannot be computed for $x > 80$.

(D) It is reliable, because the model is linear.

1. The model was fitted for x from 20 to 80, and the prediction is for $x = 150$.
2. Using a model outside the range of its data is extrapolation. The linear pattern may not continue.
3. The calculation is possible ($0.85(150) + 12 = 139.5$), but it is less reliable because 150 is outside the range of the data.

Answer: (B)

44. Medium · No calculator needed

The function f is defined by $f(x) = (x - 1)(x - 5)$. The graph of g is the reflection of the graph of f across the y-axis, shifted up 4 units. What is the value of $g(2)$?

1. Reflection across the y-axis replaces x by $-x$, and shifting up 4 adds 4: $g(x) = f(-x) + 4$.
2. Then $g(2) = f(-2) + 4$, and $f(-2) = (-2 - 1)(-2 - 5) = (-3)(-7) = 21$.
3. $g(2) = 21 + 4 = 25$.

Answer: 25

45. Medium · No calculator needed

Two similar triangles have areas 18 and 50. The altitude to the longest side of the larger triangle is 15. What is the corresponding altitude of the smaller triangle?

(A) 10.8 (B) 5.4

(C) 12.5 **(D) 9 ✓**

1. The ratio of areas is $\frac{18}{50} = \frac{9}{25}$, so the ratio of corresponding lengths is $\frac{3}{5}$.

2. Corresponding altitudes are corresponding lengths, so they have the same ratio.

3. Smaller altitude: $15 \times \frac{3}{5} = 9$.

Answer: (D)